

Prof. Ali Mostafavi, Ph.D., PE
Zachry Endowed Professor, Director, UrbanResilience.AI Lab
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Statement of Interest — Post-Doctoral Research Scientist, UrbanResilience.AI Lab

Dear Professor Mostafavi,

I am writing to express my strong interest in the Post-Doctoral Research Scientist position. As a **Texas A&M Ph.D. graduate** (4.0 GPA) with **470+ citations** across 16 publications and a current **Senior AI/ML Engineer at Ford** building production-scale multi-agent AI and transformer systems, I am eager to bring my full expertise in AI, multi-modal learning, and large-scale data systems to the mission of **disrupting disasters with AI**. I am in the US and available immediately.

Direct Alignment with Lab Priorities. My research and industry work map directly onto the technical areas your lab prioritizes:

- **Reasoning Models & Agentic AI / Tabular RAG:** At Ford, I architected a **production multi-agent system** (LangChain, LangGraph) with chain-of-thought reasoning, **RAG over tabular data** in BigQuery, and vector databases (FAISS, ChromaDB)—an SQL Agent, Document Agent, and LLM Decider serving 10 US states. This architecture transfers directly to building disaster reasoning agents over infrastructure inventories, damage assessments, and census data.
- **Multi-Modal Learning:** My Ph.D. centered on multi-modal sensor fusion. I designed a **dual-branch transformer** with cross-attention fusion achieving **AUC 0.993** for wearable stress detection, and built **OpenSync** (31 citations)—an open-source platform synchronizing EEG, eye-tracking, and biosensor streams in real time. This extends naturally to fusing satellite imagery, social media, sensor networks, and geospatial data for disaster assessment.
- **Spatial ML & Vision-Language Models:** I co-authored a **county-level COVID-19 risk study** (9 citations) combining spatial, demographic, and socioeconomic features with ML, and deployed a **computer vision system** for pandemic safety monitoring at 98% accuracy over 50K+ daily interactions (76 citations). My combined CV and NLP depth positions me to develop vision-language pipelines for post-disaster damage assessment.
- **Graph Foundation Models:** My experience with graph-structured reasoning in agentic AI, network-level data pipelines at scale (10+ TB), and transformer architectures provides a strong foundation for developing GNNs and graph foundation models to analyze interdependent infrastructure networks and predict cascading failures.

Research Vision. I envision two high-impact contributions: **(1)** building an **agentic “disaster co-pilot”**—a multi-agent system that ingests real-time multi-modal data (satellite imagery, IoT sensors, social media, infrastructure status) and provides chain-of-thought reasoning with explainable recommendations to emergency managers; and **(2)** developing **multi-modal foundation models for urban resilience** that jointly learn from tabular infrastructure data, geospatial imagery, and textual incident reports, enabling few-shot transfer across disaster types.

Why Me. I bring a rare combination of *research impact* (470+ citations, first-authored papers in JMIR, IEEE, SN Computer Science, J. Neuroscience Methods) and *production engineering* (deployed AI systems at Ford and CereVu Medical serving real users at scale). I contributed to **VR-based disaster risk interventions** (iScience, 20 citations) and have deep experience with every technical area listed in your posting. As a proud Aggie, I would be honored to return to Texas A&M and dedicate my next chapter to building cutting-edge AI solutions with real impact on community resilience.

Sincerely, Moein Razavi, Ph.D.